



The role of Village Health Workers and challenges faced in providing primary health care in Mutoko and Mudzi Districts in Zimbabwe

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Abstract

Shortage of health staff is a global problem and this is more pronounced in Sub Saharan Africa. Realizing this challenge, World Health Organization recommended the use of Community Health Workers to provide primary health care in these countries with weak health systems and high demand for health services (WHO 2008). This study therefore focused on the role of the Village Health Workers (VHWs) and challenges faced in providing primary health care in Zimbabwe. An exploratory qualitative research study was conducted in two districts using focus group discussions, in-depth and key informant interviews, and observations. The study revealed that after undergoing a formal training VHWs performed the following roles; preventative, promotional, surveillance, referral and supportive. Challenges associated with inadequate support of VHWs as a result of limited resources were noted. These challenges however had implications on VHW programme's efficacy in the country.

Key words: primary health, village health workers, Zimbabwe.

1.0 Introduction

Globally, Community Health Workers play a significant role in the health systems of their countries. Despite their important role in providing primary health care, little focus has been provided on exploring their operations. This study therefore delved on the role of Community Health Workers and challenges they face in providing primary health care in Zimbabwe.

2.0 Background

Shortage of health professionals is a problem globally; a total of four million health professionals are needed to have all positions filled (Global Health Workforce Alliance and WHO 2011). The problem is however more pronounced in developing countries, with Sub Saharan Africa having only 3% of the global health providers (Atun et, al. 2011). Staff shortages have weakened health systems of these countries resulting in high morbidity and mortality; and this has contributed to failure by these countries to achieve targets set for health related Millennium Development Goals (Smith et, al. 2014).

Shortage of health personnel, coupled with HIV and AIDS, have overburdened Sub Saharan Africa given that these countries have been struggling to provide basic health care to their population (WHO 2006; WHO, UNAIDS, UNICEF 2008). Furthermore, weak health systems of these countries are failing to cope up with high demand of health care services considering that Anti Retro Therapy (ART) and HIV AIDS is changing from being a fatal disease to a chronic one (Hermann 2009). In response to staff shortages World Health Organization had recommended task shifting as a solution to that problem (WHO 2008). The major aim of task shifting is to use available human resources efficiently and the process is done through distribution of specific tasks from professional health personnel to less qualified ones who have undergone shorter training (WHO 2006). Prominence has therefore been given to the use of Community Health Workers as a way of task shifting in the low resourced Sub Sahara Africa.

Community Health Workers (CHWs) are people chosen by the community and trained to provide health promotion interventions and health related duties within their communities and have neither formal certification nor tertiary education (Nkonki et, al. 2011). Training received by CHWs is accepted within the health system but is not recognized by the tertiary education system (Lehman et, al. 2004). The community health worker concept became popular worldwide after the Alma Ata Declaration 1978 on improving primary healthcare, which was a set of strategies to achieve the WHO 1975 set target on "Health for All by the Year 2000" (Van Ginneken et, al. 2010). The conference identified CHWs as a third workforce for "Human Resources for Health" (Sein 2006), and since that declaration, CHWs work in areas that are not easy to reach worldwide (WHO 2006). Community Health Workers often work on voluntary basis and are agents of social change within primary health care realm (Lewin et, al.2010; Werner and Bower 1982).

Whilst CHWs have either complimented the work of trained health personnel, and or assumed specific responsibilities, there has been disagreements regarding their roles and how they are managed (Mwai et, al. 2014). Another view is that they should operate from primary health facilities; and others have also questioned the selection criteria and the reporting structures (Sanders 1990; Schneider et, al. 2010). Some however have advocated for their role

to be specialized (Smith et, al. 2014). Despite these debates research has illustrated the benefits of using CHWs in the health systems as follows; linking communities to formal health systems (Smith et, al. 2014), cost effectiveness considering that most of them are volunteers (Floyd et, al. 1997), and ability and more willingness to learn by local people when taught by CHWs (Johnson 2004). In understanding the role of CHWs and their outputs, emphasis should be given to how the CHWs are trained as there is a clear linkage between the training process, role clarity and output (O'Brien et, al. 2010).

Systematic reviews have provided evidence that CHWs are efficacious in improving coverage and quality of services. This includes interventions such as promoting delivery at health institutions, post natal services, early child caring, and Prevention of Mother to Child Transmission of HIV (Izugbara et, al. 2009; Ahluwalia et, al. 2003; Mushavi 2012; Smith et, al. 2014). Other roles where CHWs have been effective include screening and education activities of HIV and tuberculosis and the necessary support treatment and adherence (Lewin et, al. 2010; Wright et, al. 2004; Smith et, al. 2014), nutrition, immunization programmes, and malaria and pneumonia assessment and treatment (Lewin et, al. 2010; Mukanga et, al. 2011; Ngasala et, al. 2011, Smith et, al. 2014). Sustainability of these programmes is however threatened by CHWs attrition (Nkonki et, al. 2011).

Whilst the above evidence demonstrates the crucial roles performed by CHWs across the countries, the success of these interventions is dependent on specific conditions (Haines 2007; Zachariah 2006); including the initial recruitment and selection, motivation, standardization of protocols, supervision and support, remuneration and structure (Hermann et, al. 2009). Unsuccessful large scale CHWs programmes which were unsustainable and had poor health services were also identified (Berman 1987; Gilson 1989). This debate shows varying levels of outputs suggesting the importance of understanding specific contexts of each CHWs programme. Though CHWs help to make the work load of professional health staff more manageable (Lidikwe et, al. 2013), they are not a solution to weaker health systems and cannot replace functions of health facilities since they are just another constituent to the health system (Schneider et, al. 2010; Van Damme 2008). This evidence shows a complimentary role played by CHWs programmes within health systems and the potential they have in relationship to strength of a health system.

Evidence from a systematic reviews demonstrate varying levels of involvement of CHWs in the health systems in response to each country's specific health programme and priorities (Mwai et, al. 2014). A general evolving and expanded role of CHWs was observed when CHWs performed more tasks than those defined in their job descriptions; in Botswana and Malawi the role of Community Lay Counsellors and Health Surveillance Assistants respectively shifted beyond preventative to curative with the cadres performing tasks beyond their mandate and training (Ledikwe et, al. 2013; Smith et, al. 2014). There is however, paucity of research regarding specific activities carried out by CHWs in Zimbabwe considering changing disease patterns. Few studies have delved on the role of VHWs in Zimbabwe; Sanders (1990; 1992) and Lehmann et, al. (2004) have focused on the development of the VHW programme, its evolution and impact in primary health care. Provided with the above evidence where task shifting has impacted differently on the roles played by the CHWs over time across Sub Sahara Africa, and the absence of current information regarding CHWs in the country, the study therefore explored the role of CHWs in Zimbabwe.

Zimbabwe's health system is weakened by staff shortages through staff attrition. In a country where the health system is weak and the demand for health services is high, for instance with HIV prevalence rate at 13% and maternal mortality ratio at 960/100 000 (ZDHS 2010/11), there has been a renewed interest on the role CHWs.

There are different types of CHWs in the country; these are Village Health Workers (VHWs), Community Home Based Caregivers and Behaviour Change Community Based Facilitators. This study however focused on the role of VHWs who constituted the largest CHW's programme in the country and who are aligned to the Ministry of Health and Child Care (MoHCC). Zimbabwe is using CHWs under the 'Village Health Worker (VHW) programme'. The country adopted the programme in 1981 and through that programme VHWs were selected by the community at ward level, trained for a period of 8 weeks, before operating on part time basis within their communities (Sanders 1991; Mushavi 2012). The country had trained 7000 in 1987 with an expectation of reaching a target of 15 000 VHWs under that programme (Sanders 1991).

3.0 Methodology

The study was conducted in Mutoko and Mudzi Districts in Zimbabwe and the two districts were conveniently selected due to their remoteness from the metropolitan and their closeness to each other. Furthermore, sampling of two districts made it possible to make comparisons. A total of 10 primary health facilities out of 27 in Mutoko District, and 11 out of 28 in Mudzi District were randomly sampled to participate in the study. Following that 4 VHWs were then randomly selected from each primary health care facility's catchment area. Inclusion criteria for participants to take part in the study were as follows; VHWs who had worked for a minimum of 12 months, and VHWs who were willing to participate.

An exploratory qualitative study was conducted using focus group discussions (FGDs), in-depth interviews (IDIs) with VHWs, and Key Informant Interviews (KII) with senior health staff. Observations were also done on VHWs' activities both in the communities and at the primary health care facilities. Data collection was stopped after reaching saturation point; when no more new themes were emerging from data (Glaser and Strauss 1967). This was after conducting a total of 6 FGDs (n=12), 12 IDIs and 10 KIIs. A purposive sampling technique was used to select key informants and KIIs were conducted with District Nursing Officers (n=2) and District Community Nurses (n=2) who were responsible for the primary health facilities. Furthermore, professional health service providers (n=6) were also purposively drawn from the sampled primary health facilities. Data collected from FGDs, KIIs, and IDIs were triangulated with information collated from the districts' reports and VHW diaries.

Data were collected by 4 interviewers; 1 interviewer conducted the KIIs, another one IDIs and the remaining 2 took turns facilitating and taking notes of the FGDs. Interviewers also conducted observations of VHWs' activities in communities prior interviewing them. Digital recorders were used to record the interviews and the focus group discussions. In-depth interviews and FGDs were conducted in local *Shona* language except for the KIIs which were done

in English. Prior data analysis, audio recorded files were concurrently transcribed verbatim and translated followed by proof reading, which checked for accuracy and completeness. Proof reading was done by a separate interviewer from the one who transcribed and translated the audio file. Data analysis was done following principles of Grounded Theory (Glaser and Strauss 1967), and this was through open coding of all transcripts and field notes. After going through all transcripts and fieldnotes line by line and coding them, similar codes were then grouped together to form concepts and these concepts were consequently merged to form categories. These categories then formed the themes. This process started on the onset of data collection and continued after completion of data collection process and codes were continuously refined upon emerging of new data. The following themes emerged from the data; recruitment and selection, training, service coverage, tasks performed, task shifting, supervision and monitoring and challenges. A story was then constructed from a combination of these themes.

3.1 Ethical Considerations

Participants took part voluntarily in the study as they were asked to sign informed consent forms after reading and understanding the information sheet that explained the purpose of the study and what it involved. Questions that participants had were consequently addressed by the interviewer before participants consented. Anonymity was observed as participants' names were not used during the interviews and discussions, and their names did not appear also on the interview transcripts. The study's protocol was approved by the local research ethics body, the Medical Research Council of Zimbabwe (MRCZ) prior conducting the study.

4.0 Results

4.1 Recruitment and Selection

Village Health Workers were selected through a participatory process. Participants indicated that the VHWs positions were advertised in the community. This was followed by a selection process conducted through a community meeting chaired by traditional leadership and the local councillor. During the meetings, nominations of candidates were done before a voting exercise. Qualifications required for the positions were the ability to write and read. Through that selection process, community people chose the people whom they entrusted to carry out the health related duties. Gender of VHWs was predominantly female and main reason cited for that was that the communities perceived female VHWs as good care givers who could handle health problems more effectively.

4.2 Training

It was revealed by participants that prior qualifying to be a VHW, all candidates underwent a training process. Length of the training varied with each group; some participants reported to have undergone training for 3 months whilst others were trained for a period of up to 7 weeks. Participants also pointed out that the curriculum of the training included topics on general health care, water and sanitation, sexual reproductive health, and immunization. Other topics covered were HIV/AIDS, Anti Retro Therapy (ART), Prevention of Mother to Child Transmission (PMTCT), Malaria Case Management, chronic illness and HIV/AIDS, tuberculosis and maternal health. Though some participants reported to have attended at least a training following the initial one, others however indicated not to have attended any follow up training from the initial one.

It was noted that some VHWs were not formally trained and these were also working in the communities. Participants explained that this group of VHWs underwent an informal on job training under the mentorship of nurses at the local health facilities. In participants' view, though these VHWs performed the same preventive tasks as the formally trained ones, they were not fully recognized as their counterparts. Unlike the formally trained ones who carried out curative tasks in addition to receiving allowances and badges and uniforms, the informally trained ones were not provided with subsidies, identification materials and did not perform tasks beyond preventative ones. Low motivation was reported from this informally trained group. Participants however indicated that the informally trained VHWs were prioritized for recruitment and training in the oncoming formal recruitment, selection and training of VHWs.

4.3 Service Coverage of VHWs

Distribution and total number of VHWs varied with each district. Whilst Mutoko had 160 trained VHWs who were responsible for the 633 villages within that district, Mudzi had a single VHW in each of the 240 villages. Availability of resources to train and support the VHWs determined the number of cadres trained as VHWs, for instance Mudzi District benefitted from additional support from a local non governmental organization unlike Mutoko District that relied solely on support from the Global Fund. Though each village had a VHW in Mudzi District, sustainability of the programme was threatened by lack of resources to continue supporting the additional trained VHWs previously supported by a non governmental organization. For the two districts, VHWs walked or cycled to their clients' home and in rare cases attended their clients from their home. During selected days some VHWs operated from the local health centres.

4.4 Tasks Performed by Village Health Workers

Village Health Workers performed a wide range of activities within the communities they live. Tasks conducted by VHWs include educational, advocacy, curative and surveillance in both districts.

Preventative and Promotional Role

Core duties of VHWs were to perform primary health care activities through disease prevention. Participants were of the view that this was done through health educational information imparted to communities. *"My role is to encourage villagers to prevent disease outbreaks, maintaining hygienic conditions in the households, and offering health education to the general population. We also concentrate on following those [people] who are hard to reach"* (IDI; VHW). Village Health Workers also identify people with diseases before referring them to the health facility in addition to identifying and referring pregnant women to visit the health centres for ante natal care. Furthermore, VHWs were

reported to conducting eye screening and identifying malnutrition cases before referring them to the local primary health care facilities. Participants further stated that VHWs played a promotional role through marketing the health interventions. *“When the Ministry [MoHCC] has a new health programme, VHWs are key in raising awareness thereby increasing acceptability of the programme, for example when there is an immunization programme to be launched,...they raise awareness before the programme starts”* (Key Informant ; District Community Nurse).

Curative Role

Participants indicated that VHWs performed curative tasks within their communities through providing malaria testing and treatment services to clients and referring clients to the local health centres when necessary. Treating minor ailments such as headaches and stomachs and offering first aid were some of the curative tasks performed by VHWs. Though VHWs performed various activities, attending to malaria cases constituted a larger proportion of their work in comparison with other responsibilities in the two districts. It was noted that the curative role emerged in response to the high malaria burden in the two districts. *“After witnessing high morbidity and mortality due to malaria, the curative component was added on Malaria Case Management as community people were delaying visiting health centres after getting Malaria”* (Key Informant; DNO).

Surveillance Role

Through a surveillance role, it was observed that VHWs oversaw the general health situation in the villages through identifying any diseases occurring within the communities. Participants were of the view that general trends of disease types, unusual health incidents and disease trends were all recorded before being reported to the local clinic. It was through VHWs day to day interaction with communities that health information was obtained and communicated to the health service providers at local health facilities. *“They [VHWs] also report when Malaria cases are increasing and when the number of people dying in the communities due to certain diseases is increasing”* (KII; Nursing Staff).

Supportive Role

A supportive role carried out by VHWs was noted when they assisted the chronically ill patients including hypertension, diabetic, and ART clients. Participants stated that they offered necessary psychosocial support services in form of counselling and ensuring that clients took their medicines continuously. Village Health Workers were reported to be also active in tracking TB patients and observing if they were taking their medicines as prescribed.

Referral role

Another task carried out by VHWs was to refer clients seeking various health services to the local health centre, for example they screened for eye cataracts and referred clients with eye problems to health facilities. Follow up visits were also made to ensure that the referred clients actually received health services. This was done through enquiring from the client, the clients' family or checking with staff from the health facility if the client has actually sought services or not. Furthermore, VHWs were noted to refer clients to local programmes and support groups, for instance participants mentioned that some clients were referred to nutrition programmes run either by government or nongovernmental organizations within their communities whilst those on ART were referred to People Living with HIV AIDS groups. In turn, VHWs revealed that they also attended to clients referred by other organizations, for instance those suspected to be suffering from Malaria.

4.5 Tasks Shifting; Additional tasks carried by VHWs

Participants showed mixed experiences regarding performance of extra tasks from those listed on their job description; some participants from one of the districts pointed out to have carried out tasks beyond their mandate which were not listed on their job descriptions, and tasks which they did not receive a formal training for. In a few health facilities in both districts it was observed that VHWs assumed the role of nurse aids through making initial observations, documentations, recording temperatures and weighing. For these facilities VHWs took turns to visit the clinic for these tasks. For some health facilities however VHWs assisted with dispensing Opportunistic Infections medicines in addition to carrying out routine health education session before clients were attended to. *“This was not part of our job description...but now the DNO has given us an instruction to work at the clinic whenever staff from the clinic is overwhelmed by clients. We take turns to work at the health centre to offer services”* (FGD; VHWs).

Though participants indicated that they received informal on job training in handling these additional tasks, they showed some disgruntlement over that. *“Those tasks [dispensing OI drugs] are supposed to be carried out by the nurses but they are giving us pressure unnecessarily since we are required to work here at the clinic twice or thrice a week yet we have other tasks in our communities”* (FGD; VHWs).

4.6 Supervision and Monitoring

Village Health Workers worked under the supervision of nurses at the local health facilities. It was indicated that local nursing staff and the District Community Nurses conducted routine field support visits to VHWs. As part of a monitoring process, VHWs were required to submit weekly and monthly reports of their activities to the nursing staff at the local health facility, which were then submitted to the district health office. It was however stated that some VHWs were not submitting these reports as a result of disgruntlement from either bad relations between them and the clinic staff, or when they were not receiving their subsidies in time.

Village Health Workers were accountable to the community through ensuring that communities were satisfied with their performance. Whilst the activities of the VHWs were monitored by the clinic staff, a Health Centre Committee also oversaw the performance of VHWs. Participants revealed that VHWs risked being rejected by the community when they performed below expected standards. *“Sometimes a VHW is rejected by the community. This can be after misbehaving, breaching confidentiality and performing poorly”* (KII; District Community Nurse).

4.7 Challenges

Allowances

Village Health Workers are volunteers who are provided with a subsidy of USD42 per quarter. Participants indicated that these allowances were meagre considering that they spent the majority of their time performing health related duties. Furthermore, there were inconsistencies on when VHWs received these allowances as some mentioned that they had not received any of their allowances for the past eighteen months. *“We are subsistent farmers and we do not get time to plant our fields as we are occupied with offering health services. We also understand that we are volunteers and that we are entitled to receiving allowances but we are not receiving them. This is so discouraging since we also need that little cash for our survival”* (Female Participant; IDI).

Lack of and, or delay of allowances has negatively affected motivation of VHWs. *“ [Delay in receiving allowances].... is the reason for some VHWs resigning from their positions. It is also unclear to whom one should enquire when VHWs delay receiving these allowances”* (Key Informant; District Community Nurse). Lack of motivation has influenced some VHWs to trade off their health care roles with tending for their families; for example some VHWs were noted to have assumed other community positions such as being a political councillor whilst others have migrated to other places in search for greener pastures. When probed on how they were managing their time, others reported that they attended to their clients in the mornings and evenings, and work in their fields during the day. Lack of a transparent system on issuing VHWs with subsidies has made supervision of VHWs difficult as it was noted that some VHWs were not consistently reporting their activities to the local nurses.

Inadequate of supplies

Shortage of resources needed by VHWs to execute their duties was another challenge reported by participants. Inadequate and inconsistent supplies of stationery, malaria test kits, and drugs were negatively affecting VHWs' work. In Malaria Case Management, lack of test kits and drugs made it impossible for the VHWs to attend to clients and that left them with the only option of referring them to the nearest health centre. *“A major drawback in executing our duties is that we are told by the nursing staff from the clinic that all malaria test kits are finished. Sometimes we run short of malaria drugs. We also go for months without stationery that is needed for our records and reports”* [IDI; VHW].

Workload

Assuming many responsibilities and covering several villages by VHWs was another challenge cited by participants. Whilst in Mudzi District each village was allocated a VHW, Mutoko District had a single VHW operating in up to 8 villages. It was also revealed that being overwhelmed by workload affected their performance. Furthermore, participants reported low motivation due to excessive work load considering that VHWs were operating on voluntary basis.

5. Discussion of Major Findings

The study showed that recruitment of VHWs is conducted from within communities where the VHWs came from and this has gained them acceptance and confidence from the communities. This evidence supports the general opinion that programmes run by CHWs are efficacious in creating health awareness and health outcomes (Bang et, al. 1994; Abbatt, 2005; Lewin, 2005). This study also corresponds with earlier studies in Zimbabwe that showed VHWs being accountable to both the community and to the health system (Sanders 1992). Provided with the empowering role CHWs programmes have when they are well rooted within the communities, the VHWs programmes has the potential to address inequalities as communities are empowered to solve other local problems (Prasad and Muraleedharan 2007).

Whilst gender cohorts of VHWs are rarely reported in past studies (Lehmann et, al. 2004), this study revealed that the majority of the VHWs were females; and these findings corroborate other researches that found out that gender of CHWs was predominantly female due to the perception that female VHWs were better caregivers especially in maternal and child health (Andy et. al. 2004). Though male VHWs have had substantial role in epidemics control, it was found out that in Somalia they did not reach women particularly on a children's health programme (Bentley, 1989). The above evidence, where the VHWs programme in the country is dominated by women, shows gender dimensions within the communities and this has implications on acceptability of health interventions at community level.

Results of this study were that VHWs performed a wide range of tasks including preventative, promotional, curative, surveillance and referral. This corresponds with past studies that identified the same roles performed by CHWs throughout Sub Saharan Africa though these roles varied with each country's health priorities and needs (Lehmann et, al. 2004; Mwai et, al. 2014). No specialization of tasks performed by VHWs were reported in this study. This however contradicts findings from other researches that showed some specialization of tasks performed by VHWs. Lehmann et, al. (2004) identified some level of specialization of tasks by CHWs across Sub Saharan Africa; including malaria treatment in Zaire and Gambia, food nutrition in Madagascar, Family planning in Rwanda and Tanzania, refugee work in Tanzania, and environment and rehabilitation work in South Africa.

The study also established that VHWs performed activities beyond their mandate and some of them felt that they were being instructed to take nurses' tasks yet there was no compensation for that. Given that VHWs were volunteers who were not receiving their subsidies sufficiently and regularly and had excess workload, sustainability of the VHWs programme is threatened. Most CHWs trainings do not cover curative tasks as such it is important to balance between the preventative and curative tasks CHWs perform (Prasad and Muraleedharan 2007). These outcomes suggest that participants should be formally trained on extra activities including curative and possibly include these tasks on their job descriptions. This however has implications on subsidies given to VHWs as they will be spending more time performing the expanded tasks.

Outcomes of this study also showed that financial constraints affected sustainability of the VHWs programme. This concurs with other researches that demonstrated that CHWs required medium to long term funding for sustainability as

funding is not only required for recruitment, selection and training but for daily supplies and maintaining the staff at work (Cambell et, al. 2008). Evidence from CHWs programmes in Sub Saharan Africa demonstrates that inadequate subsidy to CHWs negatively impacts on effectiveness of CHWs' programmes (Mwai et, al 2014). These results suggest the need to address the problem of funding of CHWs in the country to ensure that they were properly trained and that high motivation of the cadre is maintained.

The study also revealed that in one of the districts some VHWs covered a wider area beyond their capacity in order for them to operate effectively. In view of that, Prasad and Muraleedharan (2007) maintained that CHWs were most likely to be more intensive and effective in their work when covering a smaller area. This therefore raises questions on programme's efficacy when each one of the VHWs covered several villages.

5.1 Recommendations for further studies

Future studies should explore the efficacy of VHWs programmes in the country in providing specific services such as HIV and ART, improving maternal health, and malaria case management.

6.0 Conclusion

Village Health Workers play an important role in providing primary health care within the two districts through performing educational, promotional, curative, and surveillance roles. Task shifting was evident in both districts when VHWs played an expanded role when they performed some curative tasks including some which they were not formally trained for. Though VHWs were directly supervised by the local clinic staff they were also accountable to their communities. Challenges faced by the VHWs were having a huge work load beyond their capacity, and inadequate and erratic support in form of subsidies and supplies from the health authorities.

Authors' Contributions

Oliver T. Gore Conceived the study, formulated the study's objectives, and developed study's protocol. Following that, he participated in data collection with co-authors, analyzed data before he drafted the manuscript.

Festus Mukanangana Participated in refinement of methodology, data collection and analysis and reviewing of the manuscript draft.

Collet Muza Participated in improvement of methodology, data collection and analysis, and reviewing of the manuscript draft.

Manase Kudzai Chiweshe Assisted in the conceptualization of the study, data analysis and reviewing of the manuscript draft.

References

1. Abbatt, F. (2005). Scaling up Health and Education Workers: Community Health Workers. London. DFID Health Systems Resource Centre.
2. Ahluwalia, I. B. Schmid, T. Kouletio, M. & Kanenda, O. (2003). An evaluation of a community-based approach to safe motherhood in northwestern Tanzania. *Int J. Obstet Gynecol.* 82(2), pp. 231-40.
3. Atun, R. Bataringaya, J. (2011). Building a durable response to HIV/AIDS: implications for health systems. *J Acquir Immune Defic Syndr.* 57 (Suppl 2), pp. S91-5.
4. Bang, A.T. Baitule, S. B. Reddy, H. M, Deshmukh, M. D. & Bang, R. A. (2005). Low Birth Weight and Preterm Neonates: can they be managed at home by mother and a trained village health worker? *Journal of Perinatology*, 25, pp. S72-S81.
5. Bentley, C. (1989). Primary health care in north western Somalia: a case study. *Soc Sci Med.* 28 (10), pp. 1019-1030.
6. Berman, P. Gwatkin, D. Burger, S. (1987). Community-based health workers: Head start or false start towards health for all? *Soc ScieMed.* 25(5), pp. 443-449.
7. Campbell, C. Gibbs, A. Maimane, S. Nair, Y. (2008). Hearing community voices: grassroots perceptions of an intervention to support health volunteers in South Africa. *Sahara J.* 5(4), pp. 162-177.
8. Gilson, L. Walt, G. Heggenhougen, K. Owuor-Omondi, L. Perera, M. Ross, D. & Salazar, L. (1989). National Community Health Worker Programs: How Can They Be Policy. 10(4), pp. 518-532.
9. Global Health Workforce Alliance and WHO (2011). Global Experience of Community Health Workers for Delivery of Health Related Millennium Development Goals: A Systematic Review, Country Case Studies, Recommendations for Integration in to National Health Systems. Geneva: World Health Organization.
10. Mwai, G. W. Mburu, G. Torpey, K. Frost, P. Ford, N. & Seeley, J. (2013). Role and outcomes of community health workers in HIV care in sub-Saharan Africa: a systematic review. *Journal of the International AIDS Society.* 16, pp. 18586.
11. Haines, A. Sanders, D. Lehmann, U. Rowe, A. Lawn, J. Jan, S. Walker, D. & Bhutta, Z (2007). Achieving child survival goals: potential contribution of community health workers. *Lancet*, 369 (2121), pp. 2121-2131.
12. Hermann, K. Van Damme, W. Pariyo, G. W. Schouten, E. Assefa, Y. Cirera, A. & Massavon, W. (2009). Community Health Workers for ART in sub-Saharan Africa: learning from experience – capitalizing on new opportunities. *Human Resources for Health*, 7:31 doi:10.1186/1478-4491-7-31.
13. Izugbara, C, Ezeh, A, & Fotso, J. C. (2009). The persistence and challenges of homebirths: perspectives of traditional birth attendants in urban Kenya. *Health Policy Plan.* 24(1) pp. 36-45.
14. Johnson, B. A. Khanna, S. K. (2004). Community health workers and home-based care programs for HIV clients. *J Natl Med Assoc.* 96(4) pp. 496-502.
15. Lewin, S. Munabi-Babigumira, S. Glenton, C. Daniels, K. Bosch-Capblanch, X. van Wyk, B. E. Odgaard-Jensen, J. Johansen, M. Aja, G. N. Zwarenstein, M. Scheel, I. B. Et, al. (2010). Lay health workers in primary and community health care for maternal and child health and the management of infectious diseases. *Cochrane Database of Systematic Reviews.* doi:10.1002/14651858.
16. Ledikwe, J. H. Kejelepula, M. Maupo, K. Sebetso, S. Thekiso M. Floyd K, Wilkinson, D. & Gilks C. (1997). Comparison of cost effectiveness of directly observed treatment (DOT) and conventionally delivered treatment for tuberculosis: experience from rural South Africa. *BMJ.* 315(7120), pp. 1407-11.
17. Johnson, B. A. & Khanna, S. K. (2004). Community health workers and home-based care programs for HIV clients. *J. Natl Med Assoc.* 96(4), pp.496-502.

18. Lehmann, U. Friedman, I. Sander, D. (2004). Review of the Utilization and Effectiveness of Community Based Health Workers in Africa.
19. Lewin, S. Dick, J. Pond, P. Zwarenstein, M. Aja, G. Wyk, B. V. et. al. (2005). Lay health workers in primary and community health care. The Cochrane Database of Systematic Reviews: John Wiley & Sons, Ltd.
20. Smith, M. Mbayi, B. Houghton, N. Thankane, K. O'Malley, G. & Semo, B. (2013). Evaluation of a Well-Established Task-Shifting Initiative: The Lay Counselor Cadre in Botswana, Volume 8 Issue 4 e61601. PLOS ONE www.plosone.org.
21. Mukanga, D. Babirye, R. Peterson, S. Pariyo, G. W. Ojiambo, G. Tibenderana, J. K. et al. (2011). Can lay community health workers be trained to use diagnostics to distinguish and treat malaria and pneumonia in children? Lessons from rural Uganda. *Trop Med Int Health*. 16(10), pp. 1234-42.
22. Mushavi, A. (2012). Zimbabwe Efforts to Scale Up and Intergrate Community Support with Medical Services to End Vertical Transmission. National PMTCT and Pediatric HIV Care and Treatment Coordinator: Symposium on Closing the Gap: Ending Vertical Transmission Through Community Action Georgetown Hotel and Conference Center 20/07/2012.
23. Ngasala, B. E. Malmberg, M. Carlsson, A. M. Ferreira, P. E. Petzold, M. G. Blessborn, D. et al. (2011). Effectiveness of artemether-lumefantrine provided by community health workers in under-five children with uncomplicated malaria in rural Tanzania: an open label prospective study. *Malaria J*. 10:64.
24. Nkonki, L. Cliff, J. & Sanders, D. (2011). Lay health worker attrition: important but often ignored *Bull World Health Organ*, 89:919–923 | doi:10.2471/BLT.11.087825.
25. Van Ginneken, N. Lewin, S. Berridge, V. (2011). The emergence of community health worker programmes in the late apartheid era in South Africa: An historical analysis 71 (2010) 1110e1118.
26. Werner, D. & Bower, B. (1982). Helping health workers learn. Hesperian Foundation.
27. Wright, J. Walley, J. Philip, A. Pushpanathan, S. Dlamini, E. Newell, J. et al. (2004). Direct observation of treatment for tuberculosis: a randomized controlled trial of community health workers versus family members. *Trop Med Int Health*. 9(5), pp. 559-65.
28. Aboubaker, S. Qazi, S. Wolfheim, C. Oyegoke, A. Bah, R. (2008). Community health workers: A crucial role in newborn health care and survival December 2014 • Vol. 4 No. 2 • 020302
29. Sanders, D. (1990). *Journal of Social Development in Africa* 5,1, pp. 5-22.
30. Schneider, H. Hlophe, H. Van Rensburg, D. (2008) Community health workers and the response to HIV/AIDS in South Africa: tensions and prospects. *Health Policy and Planning*. 23:179-187.
31. Schneider, H. & Lehmann, U. (2010). Lay health workers and HIV programmes: implications for health systems. *AIDS care*. 22 (Suppl 1), pp. 60-7.
32. Sein, U. T. (2006). Health Volunteers: Third Workforce for Health-for-All Movement. *Regional Health Forum*, vol 10 (no 1), pp. 38 - 48.
33. Siabani, S. Driscoll, T. Davidson, P. M. & Leeder, S. R. (2014). A randomized controlled trial to evaluate an educational strategy involving community health volunteers in improving self-care in patients with chronic heart failure: Rationale, design and methodology. Siabani et al. *SpringerPlus*, 3:689 <http://www.springerplus.com/content/3/1/689>.
34. Smith, S. Deveridge, A. Berman, J. Negin, J. Mwambene, N. Chingaipe, E. Lisa, M. Puchalski Ritchie, L. M. P. & Martiniuk, A. (2014). Task-shifting and prioritization: a situational analysis examining the role and experiences of community health workers in Malawi.
35. Prasad, B. M. & Muraleedharan, V. R. (2007). Community Health Workers: a review of concepts, practice and policy concerns, International Consortium for Research on Equitable Health Systems (CREHS), UK Government Department for International Development (DFID) lead by London School of Hygiene & Tropical Medicine (LSHTM), UK.
36. Smith, S. Delveridge, A. Berman, J. Negin, J. Mwambene, N. Chingaipe, E. Ritchie, L. M. P. & Martiniuk, A. (2008). *Human Resources for Health*, 12:24. <http://www.human-resources-health.com/content/12/1/24>
37. Glaser, B. G. and Strauss, A. L. (1967). The discovery of grounded theory: strategies for qualitative research. Chicago: Aldine.
38. World Health Organization (2008). Task shifting. Rational redistribution of tasks among health workforce teams: global recommendations and guidelines.
39. Van Damme, W. Kober, K. & Kegels, G. (2008). Scaling-up antiretroviral treatment in Southern African countries with HRH shortage: How will health systems adapt? *Soc Sci Med* 66(10), pp. 2108-2121
40. O'Brien, M. J. Squires, A. J. Bixby, R. A. & Larson, S. C. (2010). Role Development of Community Health Workers: An Examination of Selection and Training Processes in the Intervention Literature, *Am J Prev Med*. 2009 December; 37(6 Suppl 1): S262–S269. doi:10.1016/j.amepre.2009.08.011
41. WHO: (2006) Taking stock: Health worker shortages and the response to AIDS. Geneva: WHO.
42. WHO (2006a). The World Health Report 2006- Working together for health. The World Health Report. New Delhi: WHO.
43. Geneva: WHO. WHO, UNAIDS, UNICEF (2008). Towards Universal Access. Scaling up priority HIV/AIDS interventions in the health sector. Progress Report 2008. Geneva: WHO.
44. UNAIDS (2008) Report on the Global HIV/AIDS Epidemic 2008. Geneva: UNAIDS.
45. Zachariah R, Teck R, Buhendwa L, Labana S, Chinji C, Humblet P Harries A (2006). How can the community contribute in the fight against HIV/AIDS and tuberculosis? An example from a rural district in Malawi. *Transact Royal Soc Trop Med & Hyg*. 100, pp. 167-175.